Applying CRISP-DM to BIM Coordination Automation

Business Scenario and Objective

A medium-sized architecture and engineering (A&E) firm coordinates architectural, structural, and MEP models using Building Information Modeling (BIM). Manual clash detection and issue tracking consume substantial engineering hours, and late discoveries risk costly field rework.

Business Goal: Improve coordination efficiency and reduce late-stage design conflicts using historical project data.

Measurable Objective: Reduce the average number of unresolved coordination issues per project by 20% within six months of deployment.

Key Business Questions

1. Which MEP and structural interfaces historically generate the highest volume of unresolved clashes?
2. What model attributes or spatial parameters reliably predict high-risk coordination conflicts?
3. What rule thresholds filter out geometric noise without missing genuine coordination errors?

Applying the CRISP-DM Phases to the Business Problem

1. **Business Understanding:** Frame manual coordination bottlenecks as a classification/filtering problem and establish the 20% reduction baseline and false-positive tolerance.
2. **Data Understanding:** Audit historical clash matrices, BCF issue logs, and model revision metadata to assess data completeness and label reliability.
3. **Data Preparation:** Clean duplicate clashes, standardize issue classifications across disciplines, and engineer features (e.g., spatial density, discipline pairs).
4. **Modeling:** Build automated clash-filtering algorithms and predictive models to score and prioritize high-risk coordination issues.
5. **Evaluation:** Validate model predictions against senior coordinator reviews to verify precision, reduction in manual review hours, and alignment with the 20% target.
6. **Deployment:** Integrate priority-scoring scripts into weekly BIM review workflows and track resolution rates across active projects.

Expected Business Value

The project could reduce repetitive review work, identify recurring coordination problems earlier, and improve project efficiency. Human reviewers would still validate flagged issues because not every geometric intersection represents a genuine design error.